

Fundamentals of Robotics

Introduction

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What are robots

- Robot is a Czech word, *robota*, meaning "forced labour"
- The word was introduced by a novelist, Karel Čapek, in 1920 and became famous for the vast body of Science Fiction that it generated
- More recently the word refers to a well defined technological domain, which over the course of the last years, has generated a multi-billion euros worth market

A definition

- The Longman dictionary entry for robot is

A machine that can move and do some of the work of a person
and is usually controlled by a computer

- The stress is on the ability to move and on the system intelligence

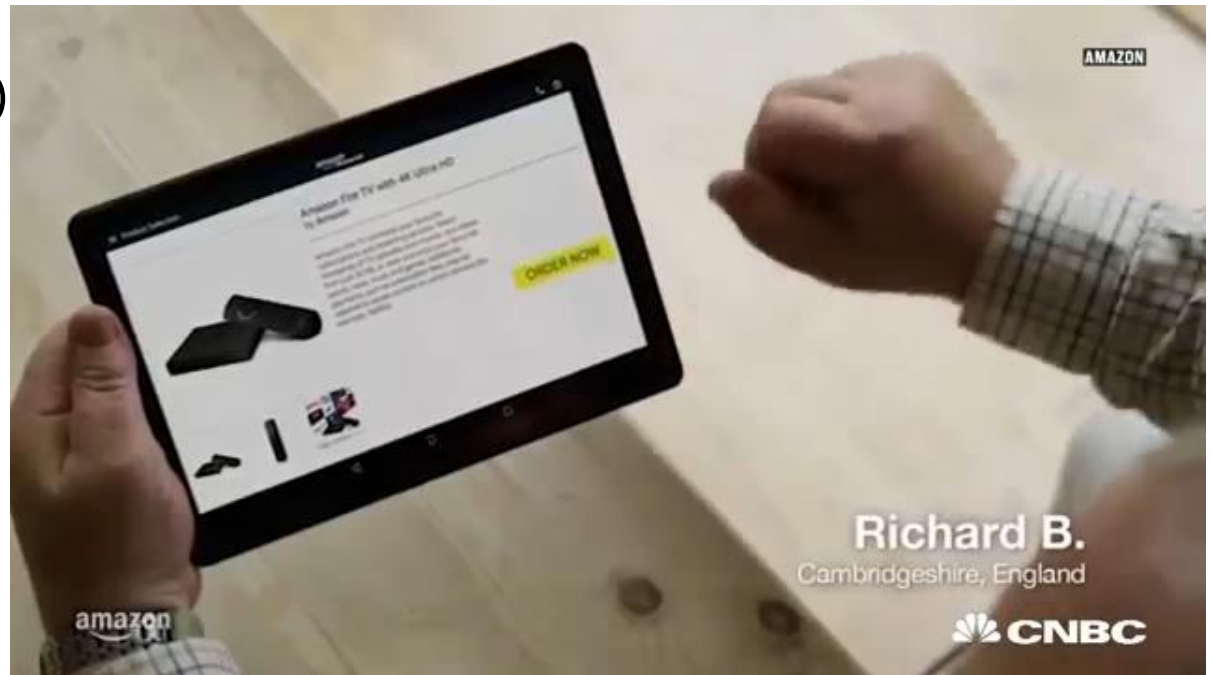
Example 1

- Robotic Arm
 - See <https://www.youtube.com/watch?v=y3e1P-OMq0Q>
 - It moves to operate on the manufactured piece
 - Its motion is controlled at different levels
 - Local: motion controllers on the joints
 - Global: motion plan to execute



Example 2

- Drone
 - see <https://www.youtube.com/watch?v=MR9PoBAssw0>
 - It can fly (autonomously or controlled) for surveillance, delivery or just for fun
- Control levels
 - Motors (rpm of each motor)
 - Trajectory following (on board)
 - Mission planning (remote)



Example 3

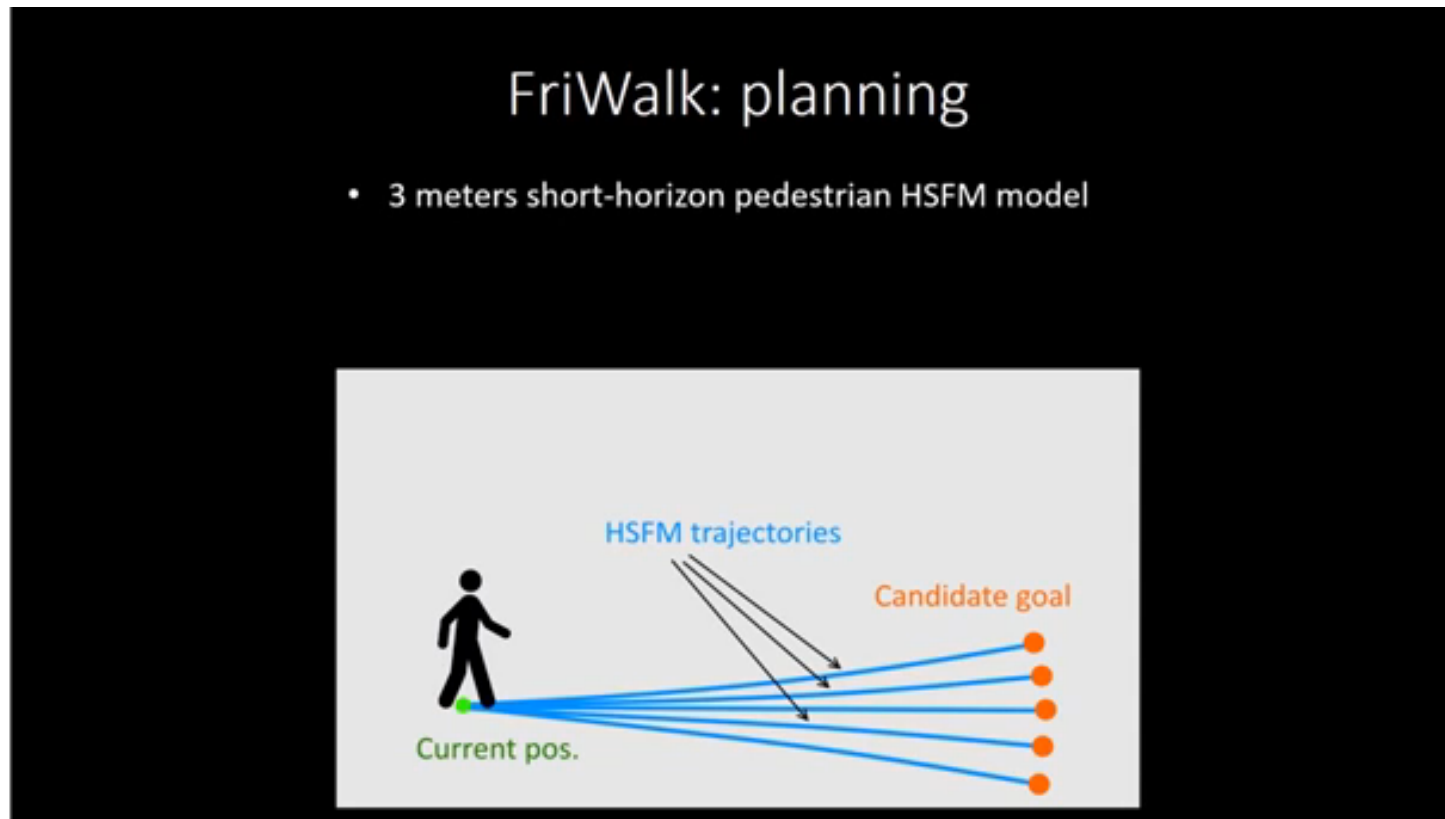
- Food Delivery robot
- See <https://www.youtube.com/watch?v=cZTCmx6N7Xc>
- Control levels
 - Motors
 - Path
 - Mission



Example 4

- A quick look to “made in Trento” assistive technology

<https://www.youtube.com/watch?v=haYAwUh68mI>



A few observations

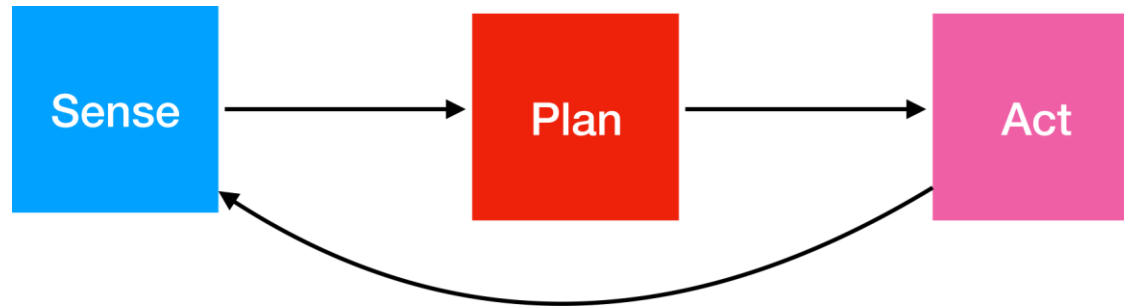
- The robots shown in the previous examples have the following abilities
 - They estimate their position (more generally their state) in the environment
 - They can sense the environment and create a map
 - They take a decision
 - They act to implement the decision

The system intelligence

- The system intelligence is concentrated in two different aspects
 - **Perception** (surrounding environment, and ...itself in the environment)
 - **Decision** (what to do and how to do it)
- Our course will revolve around these two aspects

Architectures

Deliberative Paradigm



- Top/Down philosophy
- The Robot

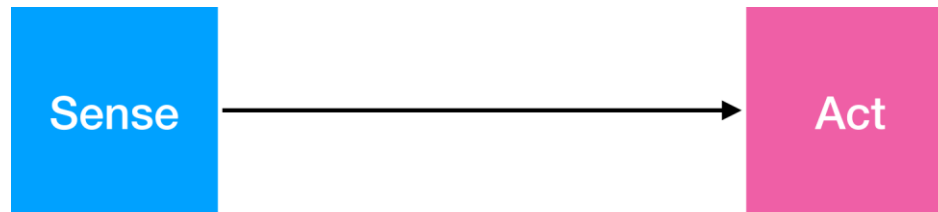
1. Senses the world and constructs an updated model of the environment
2. Plans a course of action
3. Implements the plan

Assumption: The environment is closed and its only changes are due to the robot action

Deliberative Paradigm

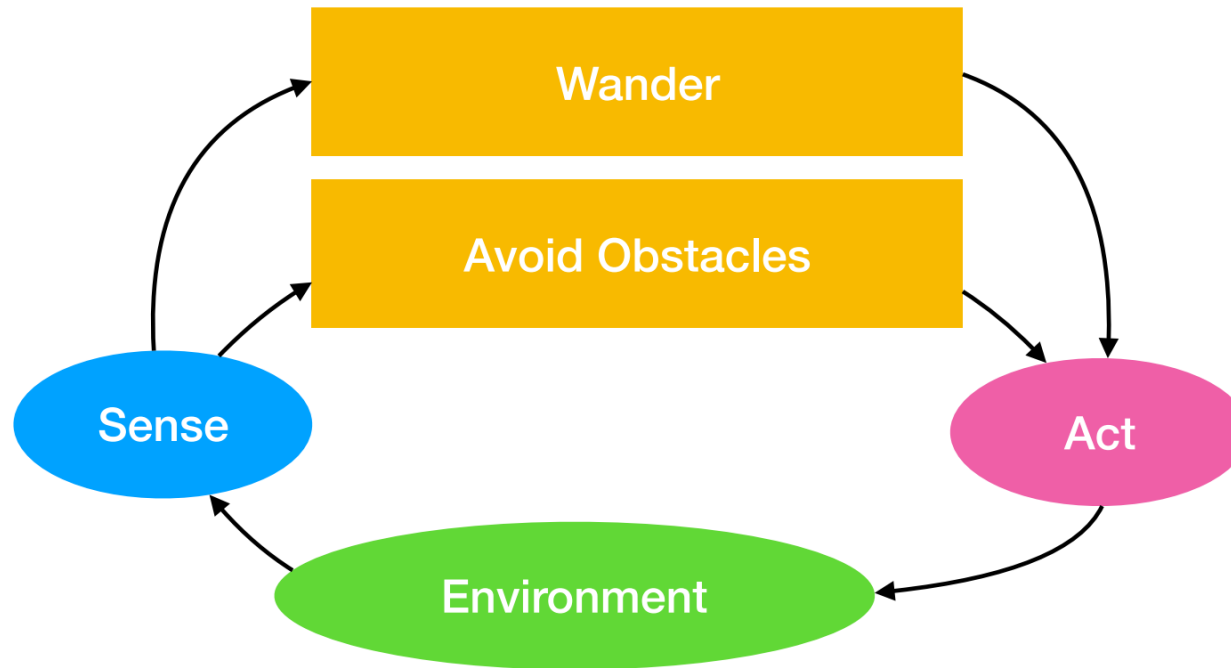
- In this paradigm all decisions are taken after a thorough deliberation procedures
- Planning is prominent because we will not make corrections before the next sensing operation
- We need a comprehensive model of the environment
- The responses of the system are therefore quite slow

Reactive Paradigm



- Sens-Act organisation
- A reaction in response to a sensed data
- Several Sensing/Reaction pairs are stored (system behaviours), which operate concurrently
- The robot will actually execute a combination of behaviours

Reactive Paradigm



- **Situated** agent, robot is integral part of the world.
- **No memory**, controlled by what is happening in the world.
- **Tight coupling** between perception and action via behaviours.
- Only local, behaviour-specific sensing is permitted (**ego-centric** representation).
- The overall behaviour of the robot emerges from the combination of the different behaviours

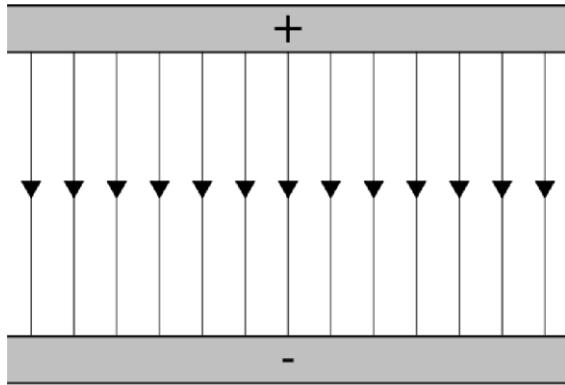
Example Potential Fields

- Treat the robot as a **particle** acting under the influence of a potential fields
- The Robot travels along the derivative of the potential
- The field depends on obstacles, desired travel directions and targets
- The Resulting field (vector) is given by the **summation of primitive fields**
- The strength of the field may change with distance to obstacle/target

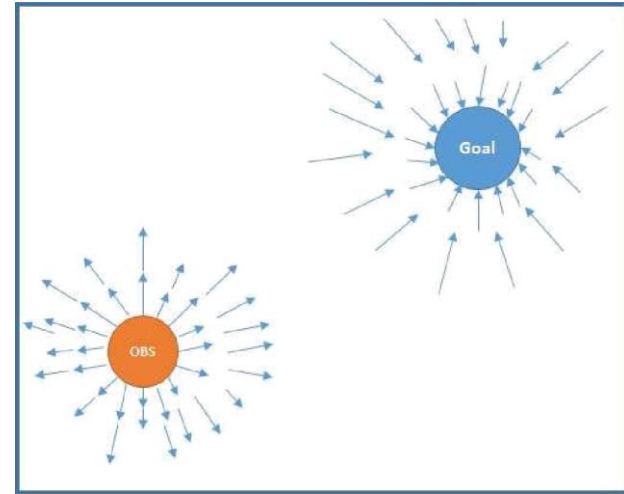
Example Corridor Following

- **Level 0** (collision avoidance) is done by the repulsive fields of detected obstacles.
- **Level 1** (wander) adds a uniform field.
- **Level 2** (corridor following) replaces the wander field by three fields (two perpendicular, one uniform).

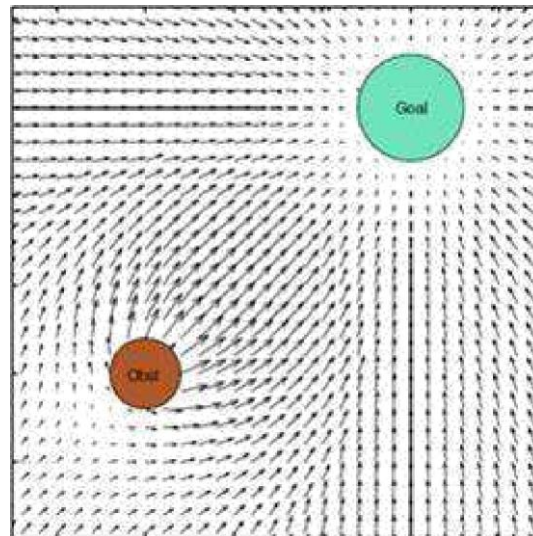
Potential Fields



Uniform Potential Field



Attractive/Repulsive Fields



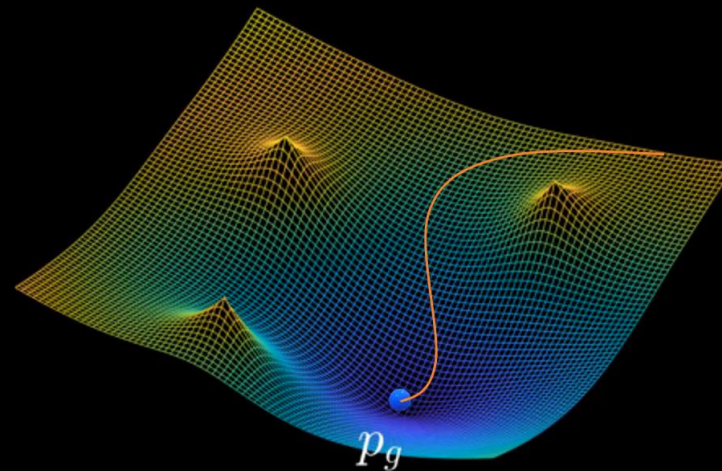
Combination

An Example

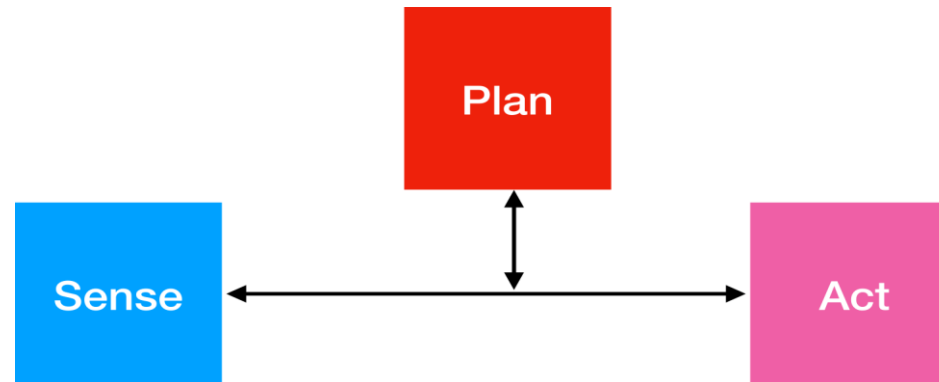
Navigation via Potential Fields for Nonholonomic vehicles

Since we assume to know the environment, we compute a safe path before the guidance starts.

By discretising it in way-points and by changing the potential function according to the active waypoint, we are able to avoid local minima generated by static obstacles in the map.



Hybrid Paradigm



- The robot first plans (deliberates) how to best decompose a task into subtasks (also called “mission planning”) and then what are the suitable behaviours to accomplish each subtask.
- Then the behaviours starts executing as per the Reactive Paradigm.
- Sensing organisation is also a mixture of Hierarchical and Reactive styles; sensor data gets routed to each behaviour that needs that sensor, but is also available to the planner for construction of a task-oriented global world model.

Hybrid Paradigm

- Hybrid Paradigms combine a good attention for planning with the ability for the system to react to unexpected changes in the environment
- The idea is to have combination of strategy (for planning) and tactics (for the execution of subtasks)
- The system maximises performance paying the price of a high level of complexity.

Perception

- In the context described above perception is key
- ...but what is perception?

Perception is a cognitive activity that makes the system aware of itself and of its environment

- It helps the robot respond such queries as

How can I detect and classify the different entities of my surrounding environment

How can I construct a map that contains both metric and semantic information?

Where am I in the map right now?

Where are the objects I need to interact with or avoid?

- This information is key to take the right decisions both in terms of which action I should do and of which path I should follow.

Perceiving humans

- Humans are increasingly part of the robot's horizon
- Sometimes the robot is required to operate in an environment populated by humans and has to preserve their safe
- Sometimes the robot is supposed to collaborate with the human (e.g., cobots)
- Sometimes the robot is supposed to assist the human
-Once again a reliable perception is key

Where are the humans in the scene
and where are they going?
(SAFETY)

What action is a human doing right
now, and what will she/he do in the
near future?
(COLLABORATION)

What is the human 's affective state
right now?
(ASSISTANCE)

- In the course we will provide an overview on the main techniques used for robot perception and will put our hands in the dirt with some of them.

Challenges

....in order to identify the architectural solution we need a clear understanding of the challenges before us.....

Challenge #1

Free robots for a free world

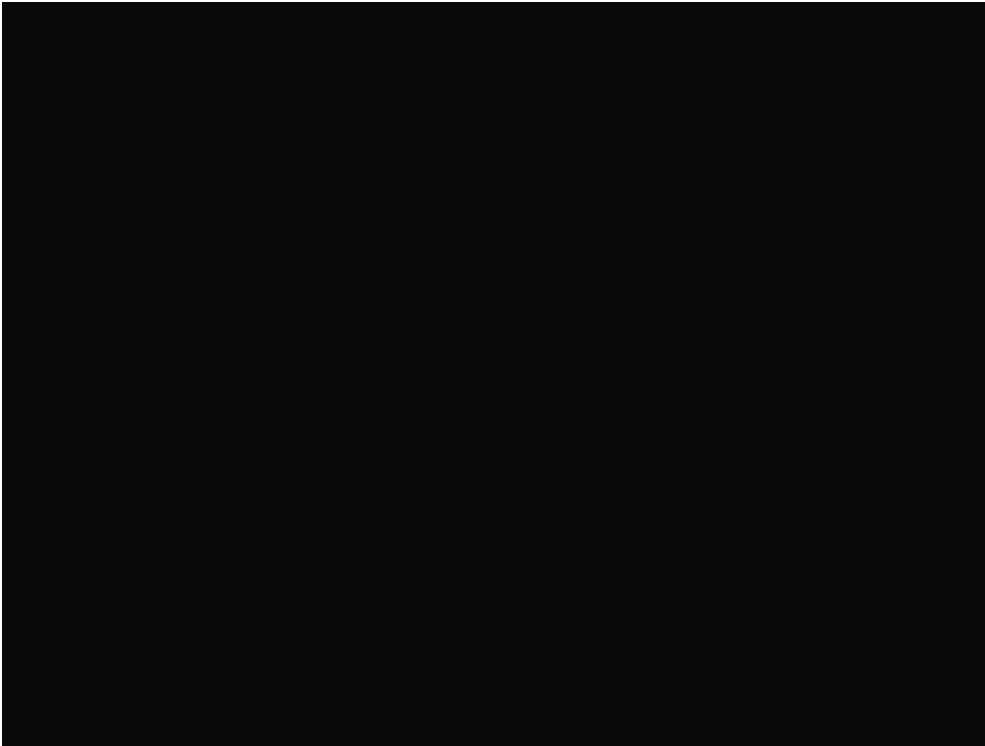


A new generation of robots operates in unknown and hostile environments.

Perception and deliberation will be intertwined: high level plans will have to be refined adapting them to the environment.

Challenge #2

Versatility



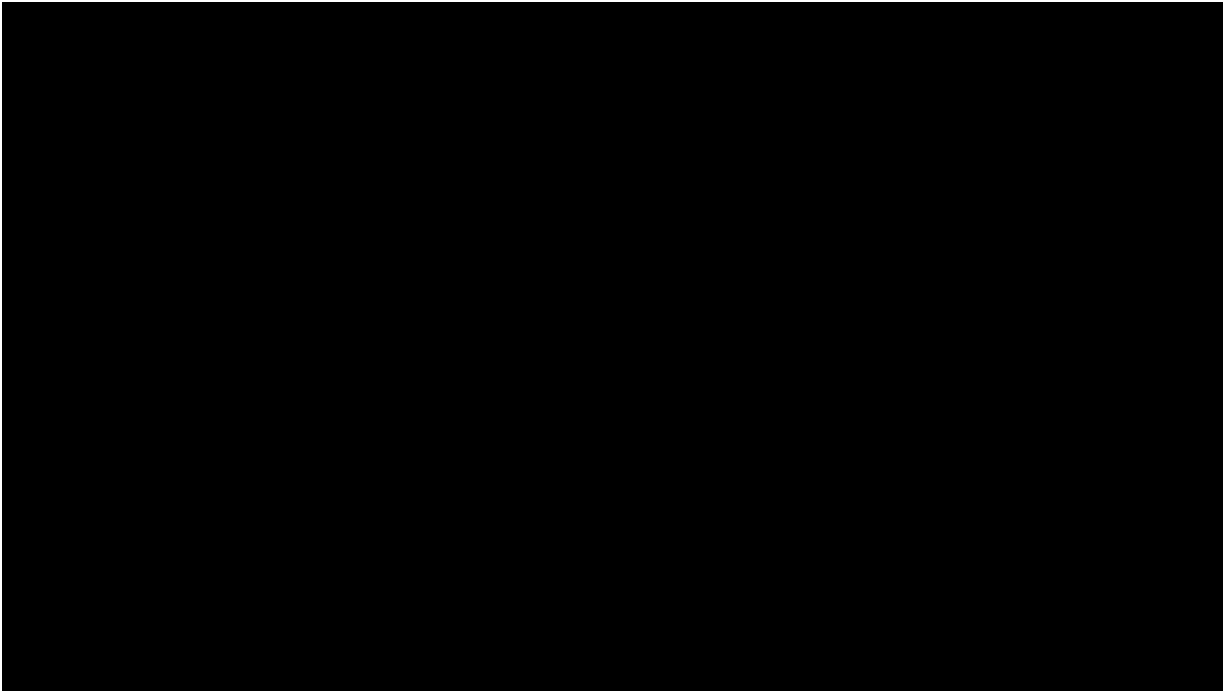
<http://personalrobotics.stanford.edu/>

Sometimes the robot has to discover its mission on its own or find non trivial ways of fulfilling its tasks

Deliberation can become a complex cognitive activity

Challenge #3

Reactivity



<https://www.youtube.com/watch?v=gsfkGISajHQ>

Sometimes the loop has to be closed very quickly

Efficiency in the use of computation resources becomes an issue

Challenge #4

Sense and sensitivity



<https://www.youtube.com/watch?v=8e76BjH9ez4>

<https://www.youtube.com/watch?v=Sq36J9pNaEo>



The robot is often called to understand its human partners and even produce empathic reactions

Sensing and understanding humans is key

Course contents

....almost time to peek into the course contents...

Foreword

1. The course adopts a very practical approach
 - You will receive an introduction to the most important theoretical foundations
 - Most of the learning effort will be by concrete programming examples and lab experiences
2. You will learn to program complex robotic systems using state of the art frameworks (the ROS middleware)
3. You will develop (and be graded on) a project, in which you will cover different aspects of robotics
 - You can work in groups (max 3 people) but the grading is individual
4. You are required to use C++ for the delivered code.
Additionally, you can use Python/Matlab for fast prototyping

Course contents

TOPIC	HOURS	Teacher
Introduction to the course	2	Palopoli
Programming Tools for robotic applications • The ROS middleware • The Gazebo Simulation Environment • The matlab/Simulink toolsuite	6	Palopoli
Project Presentation and group composition	2	Palopoli
Physical models for robots • Geometric Models • Kinematic Models • Dynamic Models	8	Palopoli
Simulation of Robot Motion	4	Palopoli
Sensing Devices • Linear Encoders/IMU • RGB cameras • Lidars • Stereoscopic cameras	2	Sebe

Course contents

TOPIC	HOURS	Teacher
Acquiring and filtering information from sensing devices • Camera Calibration • Image Rectification	2	Sebe
Classifying objects and localizing them in space • Examples with Open Cv/ Yolo	4	Sebe
Localisation/Mapping/Slam	4	Sebe
Detecting Humans • Classifying humans • Estimating Pose • Recognising gestures/affective states	6	Sebe
Sensor Fusion • Kalman Filtering • Particle Filtering	2	Sebe
Lab Experiences/Work on the project	4	Sebe

Course contents

TOPIC	HOURS	Teacher
Planning • The Planning Problem	2	Palopoli
Motion Planning • Optimal Control based motion planning • Using Sample based models • Laboratory work	10	Palopoli
Task Planning • Examples of task planning	2	Palopoli
Motion Control • Basic Information on Feedback Control • Trajectory Following • Path following	14	Palopoli
Laboratory experience on the project	16	Palopoli/Sebe
Project Wrap-Up and public discussion	4	Palopoli/Sebe

Before the next class

- Before the next class we will need to have ROS installed (either on your PC or a VM).
- We will use the noetic distribution
- The installation instruction can be found here:

<http://wiki.ros.org/Installation/Ubuntu>

You can also do on a VM (with some performance issue if your PC is not powerful enough).